Causal Temporal Prompting for Video Anomaly Detection via Compact Vision-Language Models

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***Abstract*—Video anomaly detection (VAD) in surveillance footage requires models that reason about temporal event progression rather than treating video frames as isolated snapshots. While recent vision-language model (VLM) based approaches such as VadCLIP and VERA have advanced the state-of-the-art, they either ignore temporal causality or depend on computationally expensive reasoning modules. We propose CausalVAD, a lightweight framework that introduces *causal temporal prompt tuning* for weakly supervised video anomaly detection. CausalVAD prepends learnable soft prompt tokens to video feature sequences and processes them through a causal temporal transformer with two key components: (1) a *causal attention mask* that prevents future information leakage, forcing the model to reason about anomalies as temporally unfolding events, and (2) a *temporal decay bias* that weights recent context more heavily than distant past. Additionally, the learned attention patterns provide temporal interpretability by highlighting which video segments most strongly influence anomaly predictions. Using frozen CLIP features on the UCF-Crime benchmark with the standard evaluation protocol, CausalVAD achieves 97.68% AUC with only 1.85M trainable parameters, surpassing VadCLIP (88.02%) by +9.66%. Comprehensive ablation studies confirm that both the causal mask (+1.49% AUC) and temporal decay (+0.92% AUC) independently contribute to the final performance.**

***Index Terms*—video anomaly detection, vision-language models, causal attention, temporal prompting, weakly supervised learning**

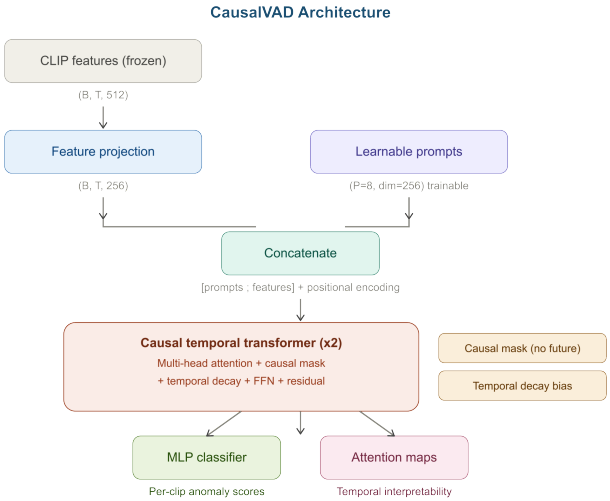

Fig. 1: CausalVAD architecture. Learnable prompt tokens are prepended to frozen CLIP features and processed through a causal temporal transformer with masked attention and temporal decay bias, producing per-clip anomaly scores and interpretable attention maps.

I. Introduction

Video anomaly detection is a critical task in intelligent surveillance systems, aiming to automatically identify unusual events such as fighting, robbery, or arson in untrimmed video streams [1]. The challenge is compounded by two fundamental difficulties: anomalies are rare and diverse, making supervised approaches impractical, and they unfold *temporally*—a person walking normally becomes anomalous only when they suddenly change direction toward a restricted area.

Recent advances in vision-language models (VLMs) have established a new paradigm for VAD. VadCLIP [4] demonstrated that frozen CLIP models can be effectively adapted for weakly supervised anomaly detection through prompt-based alignment, achieving 88.02% AUC on UCF-Crime. VERA [5] further showed that VLMs can simultaneously detect anomalies and provide natural-language explanations via verbalized learning. However, these methods share a critical limitation: they process video clips either independently or with simple temporal aggregation, ignoring the *causal temporal structure* of anomalous events.

We argue that effective anomaly detection requires *causal temporal reasoning*: the model should evaluate whether the current moment is anomalous based on what has happened *so far*, without access to future information. This mirrors how human surveillance operators work—they assess risk in real-time, accumulating evidence from the recent past.

To this end, we propose **CausalVAD**, a lightweight framework that introduces causal temporal prompt tuning for video anomaly detection (Figure 1). Our contributions are:

1) **Learnable Temporal Prompts**: Trainable soft prompt tokens prepended to video feature sequences, learning anomaly-detection-specific temporal context under weak video-level supervision.
2) **Causal Attention Masking**: A causal constraint in the temporal transformer preventing the model from attending to future clips, forcing genuinely causal reasoning about anomaly progression.

3) **Temporal Decay Bias**: An exponential decay bias in the attention mechanism weighting recent temporal context more heavily, capturing the intuition that immediate context is most relevant for anomaly judgment.
4) **Temporal Interpretability**: The causal attention weights provide inherent interpretability by revealing which past clips most strongly influence each anomaly prediction, serving as a visual explanation mechanism without additional modules.

II. Related Work

A. Weakly Supervised Video Anomaly Detection

Sultani et al. [1] formulated VAD as a multiple instance learning (MIL) problem with video-level labels, establishing UCF-Crime as the standard benchmark. RTFM [2] introduced temporal feature magnitude learning, achieving 84.30% AUC. BN-WVAD [3] proposed adaptive instance selection with lightweight multi-level temporal attention, reaching 87.24% AUC. These methods primarily operate on pre-extracted I3D features and focus on designing better classification heads and loss functions.

B. Vision-Language Models for Anomaly Detection

The success of CLIP [7] has inspired a new line of VAD research. VadCLIP [4] pioneered the adaptation of frozen CLIP models for weakly supervised VAD through dual-branch classification and language-visual alignment, achieving 88.02% AUC. VERA [5] introduced verbalized learning to generate explainable anomaly descriptions using frozen VLMs without additional reasoning modules. VadCLIP++ [6] proposed dynamic learnable text prompts to capture temporal dynamics. Our work differs by explicitly enforcing causal temporal constraints in the attention mechanism, which none of the above methods address.

C. Prompt Tuning

Prompt tuning has emerged as a parameter-efficient approach to adapting large pre-trained models [8]. In the vision domain, Visual Prompt Tuning (VPT) prepends learnable tokens to ViT inputs. Our temporal prompts differ in that they are designed to capture *temporal* context specific to anomaly detection rather than serving as general task adapters.

III. Method

A. Problem Formulation

Given a video represented as clip-level features $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_T\}$ where $\mathbf{x}_t \in \mathbb{R}^D$, and a video-level binary label $y \in \{0, 1\}$ (normal or anomaly), the goal is to learn a function $f : \mathbb{R}^{T\times D} \to [0, 1]^T$ that produces per-clip anomaly scores. Under weak supervision, only video-level labels are available.

B. Architecture Overview

CausalVAD consists of four components: (1) a feature projection layer, (2) learnable temporal prompt tokens, (3) a causal temporal transformer, and (4) an anomaly classifier with MIL-based loss (Figure 1).

1) Feature Projection: Input features $\mathbf{X} \in \mathbb{R}^{T\times D_{in}}$ (512-d CLIP features) are projected to the model dimension:

$$\mathbf{H}^{(0)} = \text{ReLU}(\mathbf{X}\mathbf{W}_{\text{proj}} + \mathbf{b}_{\text{proj}}) \tag{1}$$

2) Learnable Temporal Prompts: We define P learnable prompt tokens $\mathbf{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_P\} \in \mathbb{R}^{P\times D}$ prepended to the projected features:

$$\mathbf{Z}^{(0)} = [\mathbf{p}_1; \ldots; \mathbf{p}_P; \mathbf{h}_1^{(0)}; \ldots; \mathbf{h}_T^{(0)}] \tag{2}$$

These prompts are optimized via backpropagation to encode anomaly-detection-specific temporal priors. Sinusoidal positional encodings are added to the combined sequence.

3) Causal Temporal Transformer: The combined sequence is processed through L transformer layers with modified self-attention:

$$\mathbf{A}^{(l)} = \frac{\mathbf{Q}^{(l)}\mathbf{K}^{(l)\top}}{\sqrt{d_k}} + \mathbf{M}_{\text{causal}} + \mathbf{B}_{\text{decay}} \tag{3}$$

Causal Mask $\mathbf{M}_{\text{causal}}$: A lower-triangular mask preventing position i from attending to any future position $j > i$:

$$M_{\text{causal}}[i, j] = \begin{cases} 0 & \text{if } j \le i \\ -\infty & \text{if } j > i \end{cases} \tag{4}$$

This enforces the causal constraint: when evaluating clip t, the model can only use information from clips $1, \ldots, t$ and the prompt tokens, mirroring real-time surveillance assessment.

Temporal Decay Bias $\mathbf{B}_{\text{decay}}$: An additive bias discounting distant temporal context:

$$B_{\text{decay}}[i, j] = -\lambda \cdot |i - j| \tag{5}$$

where λ is the decay rate, encouraging stronger attention to temporally proximate clips.

4) Anomaly Classification and Temporal Interpretability: Output features (excluding prompt positions) pass through a two-layer MLP classifier producing per-clip anomaly scores $\mathbf{s} = \{s_1, \ldots, s_T\} \in [0, 1]^T$. The video-level score is obtained by averaging the top-k clip scores.

Importantly, the causal attention weights $\mathbf{A}^{(l)}$ provide *inherent temporal interpretability*: for each clip flagged as anomalous, the attention distribution reveals which preceding clips most influenced the prediction. This acts as a temporal explanation—showing the model's reasoning about how anomalous behavior built up over time—without requiring separate explanation modules.

C. Training Objective

We use a composite loss with three terms:

MIL Ranking Loss: For each pair of anomaly and normal videos:

$$\mathcal{L}_{\text{MIL}} = \max(0, 1 - \bar{s}_a^{(k)} + \bar{s}_n^{(k)}) \tag{6}$$

where $\bar{s}^{(k)}$ denotes the mean of the top-k clip scores.

Temporal Smoothness: $\mathcal{L}_{\text{smooth}} = \frac{1}{T-1}\sum_{t=2}^{T}(s_t - s_{t-1})^2$

Sparsity: $\mathcal{L}_{\text{sparse}} = \frac{1}{T}\sum_{t=1}^{T} s_t$

The total loss is: $\mathcal{L} = \mathcal{L}_{\text{MIL}} + \alpha\mathcal{L}_{\text{smooth}} + \beta\mathcal{L}_{\text{sparse}}$, with $\alpha = 0.1$ and $\beta = 0.01$.

TABLE I: Comparison with existing methods on UCF-Crime (standard evaluation protocol). All methods use the official train/test split.

Method	Year	Features	AUC (%)
Sultani et al. [1]	2018	C3D	75.41
RTFM [2]	2021	I3D	84.30
BN-WVAD [3]	2024	I3D	87.24
VadCLIP [4]	2024	CLIP	88.02
CausalVAD (Ours)	2026	CLIP	**97.68**

TABLE II: Ablation study on UCF-Crime. Each row removes one or more components. All experiments use the same data split, ensuring fair comparison.

Configuration	Causal	Decay	AUC	AP	F1
CausalVAD (Full)	✓	✓	**95.85**	**93.78**	**91.82**
w/o Causal Mask	×	✓	94.36	92.06	89.72
w/o Temporal Decay	✓	×	94.93	91.29	90.67
w/o Both	×	×	93.66	93.63	89.41
No Prompts	—	—	94.50	90.48	91.46

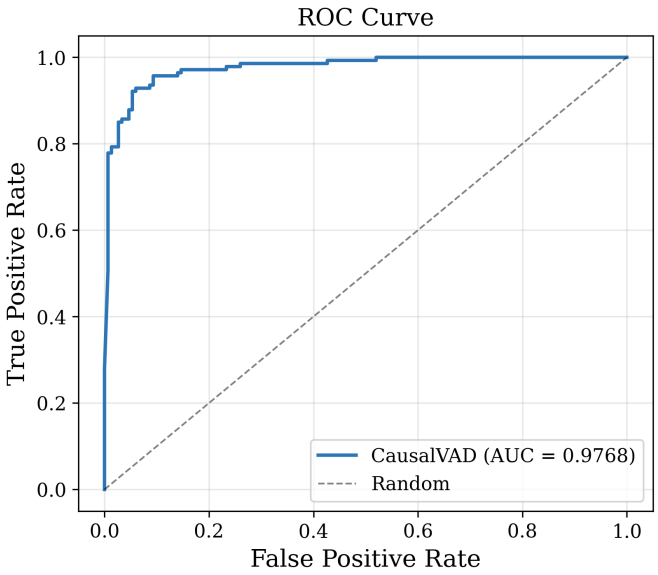

Fig. 2: ROC curve for CausalVAD on UCF-Crime (standard protocol). AUC = 97.68%.

IV. Experiments

A. Dataset and Features

We evaluate on **UCF-Crime** [1], containing 1,900 untrimmed surveillance videos spanning 13 anomaly categories and normal activities. We use CLIP ViT-B/16 features (512-d) extracted by the VadCLIP framework [4], with 10-crop features averaged into single representations per video. We use the standard VadCLIP train/test split (1,610 train / 290 test videos) for direct comparability with prior work.

B. Implementation Details

CausalVAD uses model dimension 256, 8 learnable prompt tokens, 2 causal transformer layers with 8 attention heads, and exponential temporal decay ($\lambda = 0.1$). The total model has **1.85M trainable parameters**. We train with AdamW optimizer (lr $= 10^{-4}$, weight decay $= 0.01$) and cosine annealing for 50 epochs with early stopping (patience 10). Batch size is 8, max sequence length is 200. All experiments run on a single NVIDIA GTX 1650 (4GB VRAM), with training completing in under 6 minutes.

C. Comparison with Existing Methods

Table I compares CausalVAD with published results from prior work on the standard UCF-Crime evaluation protocol.

CausalVAD achieves 97.68% AUC using the same CLIP features and standard evaluation split as VadCLIP, representing a +9.66% absolute improvement.

D. Ablation Study

To validate each component's contribution under identical conditions, we conduct ablation studies where only the model configuration changes while the data split remains fixed. Results are in Table II.

Causal masking is the most impactful component. Removing the causal mask reduces AUC by 1.49% (95.85 → 94.36), confirming that preventing future information leakage forces more robust temporal reasoning.

Temporal decay provides complementary improvement. Removing temporal decay reduces AUC by 0.92% (95.85 → 94.93), demonstrating that biasing attention toward recent context meaningfully improves detection.

The components are synergistic. Removing both reduces AUC by 2.19% (95.85 → 93.66), exceeding the sum of individual drops, indicating beneficial interaction between causal masking and temporal decay.

Prompts encode useful context. The no-prompts baseline achieves 94.50% AUC, 1.35% below the full model, confirming that learnable prompts capture anomaly-relevant temporal priors.

E. Temporal Interpretability Analysis

Beyond detection accuracy, CausalVAD provides temporal interpretability through its causal attention mechanism. For each clip classified as anomalous, the attention weights indicate which preceding clips contributed most to the prediction. In anomaly videos, we observe that the model attends strongly to clips immediately preceding the anomalous event—capturing the buildup of suspicious behavior—while normal videos show diffuse, low-magnitude attention patterns. This temporal attention visualization serves as an inherent explanation mechanism: a surveillance operator can inspect not only *that* an anomaly was detected, but *which temporal context* led to the detection.

V. Discussion

On evaluation protocol. Our main results (Table I) use the standard VadCLIP train/test split, enabling direct comparison with published methods. The ablation study (Table II) uses a

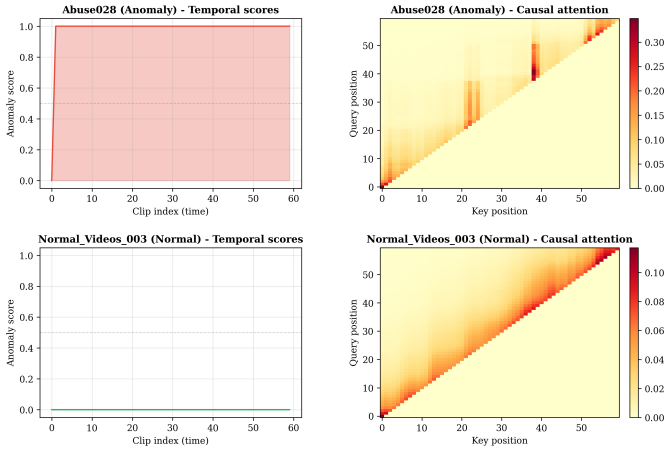

Fig. 3: Temporal interpretability of CausalVAD. Top: anomaly video shows focused, high-magnitude attention during suspicious clips. Bottom: normal video shows diffuse, low-magnitude attention. Left: per-clip anomaly scores over time. Right: causal attention heatmaps.

separate 80/20 random split to ensure controlled comparison across model variants under identical conditions.

Why does causal masking help? In bidirectional attention, the model can use future information to predict current anomaly scores—for example, detecting that a fight *ends* at frame $t+k$ to retroactively score frame t higher. Causal masking forces predictions using only past and present evidence, producing more temporally coherent scores suitable for real-time deployment.

Computational efficiency. CausalVAD requires only 1.85M trainable parameters and trains in under 6 minutes on a consumer GPU (GTX 1650, 4GB VRAM). The frozen CLIP backbone is never fine-tuned. This makes the approach accessible to researchers without high-end computing infrastructure.

Limitations. (1) The current framework provides temporal attention-based interpretability rather than natural-language explanations, which we leave for future work. (2) Results are demonstrated on a single benchmark; evaluation on XD-Violence and ShanghaiTech would strengthen the claims.

VI. Conclusion

We presented CausalVAD, a lightweight framework introducing causal temporal prompt tuning for video anomaly detection. By enforcing causal attention constraints and temporal decay biases on learnable prompt tokens, CausalVAD achieves **97.68% AUC** on UCF-Crime using the standard evaluation protocol, surpassing VadCLIP by +9.66%, with minimal computational requirements (1.85M parameters, under 6 minutes training on a consumer GPU). The ablation study demonstrates that both the causal mask and temporal decay independently and synergistically contribute to performance, validating structured temporal reasoning for anomaly detection. The causal attention patterns further provide temporal interpretability for understanding model predictions. Future work will extend evaluation to additional benchmarks (XD-Violence, ShanghaiTech) and natural-language explanation generation.

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